

Indian Institute of Technology Delhi

Civil Engineering Department

Semester I, 2026-27

Solid Mechanics and Structural Analysis Practical (CVP-2502)

Location Structural analysis lab – Block IV Room 4A6 (Third floor)

1. Lab Schedule:

Week	Thursday (Group 3)	Friday (Group 4)	Monday (Group 1)	Tuesday (Group 2)
Week 0-Introduction	23 July	24 July	27 July	28 July
Week 1	30 July	31 July	3 August	4 August
Week 2	6 August	7 August	10 August	11 August
Week 3	13 August	14 August	17 August	18 August
Week 4	20 August	21 August	24 August	25 August
Week 5	10 September	11 September	7 September	8 September
<i>Mid-Sem break</i>				
Introduction	24 September	25 September	21 September	22 September
Week 6	8 October	9 October	5 October	6 October
Week 7	15 October	16 October	12 October	13 October
Week 8	22 October	23 October	19 October	24 October (SATURDAY)
Week 9	29 October	30 October	26 October	27 October
Week 10	5 November	6 November	2 November	3 November
<i>(Common Buffer)</i>	12 November	13 November	14 November (SATURDAY)	10 November

2. **Timetable:** Lab starts at 3PM. Attendance up to 3:10 PM. **Wear shoes, if not, you'll be marked absent**, and asked to leave the lab (present medical certificate in case of any medical issue).

3. **Subgroups** - Each group will be divided in to **subgroups** at random. Each subgroup will perform one experiment in every lab class. **No change in group/subgroup will be entertained.**

4. **Lab Report:**

- **Each subgroup** will prepare a **single** lab report for every experiment.
- Each subgroup will **submit** the lab report during **the next lab class** when attendance is taken.
- All group members will be given equal marks unless someone raises any objection. A student if they wish may submit an individual report.
- Lab reports won't be accepted after the attendance is taken. If you are absent, you can't submit a report.
- Submit the reports in A4 sheet (**Hand-written**). The general content of a well written report should be as follows – Experiment Title, Objective(s), Apparatus, Diagram, Theory, Procedure, Observations/Results, and Discussions.

5. Viva

- Each student in a subgroup will have to individually give a viva for the day's experiment.

6. Grading:

- Experiments 75%
 - Report
 - Viva
- Quiz 25%.
- Grading for each Experiment will be done out of 10 Marks
 - 4 marks have been assigned to attendance and viva of experiments
 - 6 marks for the report
 - 1 mark for Aim/Objective, Apparatus and procedure
 - 1 mark for diagrams (A nicely made diagrams carry 1 mark, otherwise 0)
 - 1 marks for Theory/Derivations
 - 2 marks for Observations/ sample calculations/graph/Results
 - 1 mark for discussion.

Report Instructions

- Partial coverage of any section of the experiment will attract partial marks.
- Tables should be neatly written with units correctly indicated and labeled properly.
- Graphs should be neat. The student should write what the graph is about. Axes should be labeled.
- Percentage error should be shown and the formula for calculation of the error should be clearly indicated either in the column label itself or just before the table.
- Least counts of dial gauges and Vernier calipers, measurements of components etc. should be clearly indicated.
- One sample calculation should be shown, and rest can be tabulated. The student should be able to write clearly what method he/she is using for the analytical calculations (e.g., moment area/slope deflection method, joint method for trusses, Euler's buckling load formula etc.).
- More weightage is given to the quality of the conclusion/discussion section. Answers like "randomness of the experiment", "human errors" without indicating what they mean will be awarded zero. A student should be able to write physical explanations of the possible causes of discrepancy between the experiment and theory.
- Incomplete tables/graphs, wrong formulas, copied contents will be awarded zero.

Sample Report

Experiment 1: _____

Objective(s):

a.

b.

Apparatus:

Name	Accuracy/Least count

Diagram(s):

Theory:

Procedure:

Observations:

Calculations, Graphs, Results:

Discussions:

9. List of Experiments:

No.	Experiment	Objectives
1.	Bending Moment in Simply Supported Beam	a) To verify the principle of superposition by evaluating the bending moment in a simply supported beam due to two-point loading. b) To determine the bending moment in a simply supported beam due to two-point loading experimentally and verify the same with theoretically.
2.	Evaluation of forces and displacements in a determinate truss	a) To determine the forces in the members of a pin joined determinate truss experimentally and verify the same with theoretical values. b) To determine the horizontal and vertical deflection of various joints of a pin joined determinate truss experimentally & verify the same with theoretical values.
3.	Cable Geometry and its statics	a) To study the cable geometry and statics for different loading conditions. b) To compare the experimental and theoretical results of the horizontal reaction force for a given system of loads.
4.	Elastic deflection of simply supported beam	a) To find the value of flexural stiffness (EI) for a given beam and compare it with theoretical values. b) To verify the Maxwell's Reciprocal Theorem
5.	Uniaxial Tension test	a) To determine the ultimate axial stress, strain, and total deformation behaviour of a tensile specimen under uniaxial tension. b) To determine the elastic properties of the material
6.	Verification of Moment area theorems	To verify the moment area theorems regarding the slopes and deflection of the beam.
7.	Analysis of arch	a) To determine the horizontal displacement of one of the ends of a two hinged arch for a given load experimentally and verify the same with theoretical values. b) To obtain influence line diagram for horizontal thrust in a two hinged arch by using the Müller-Breslau's principle experimentally & verify the same with theoretical values.
8.	Analysis of a portal frame	a) To determine the bending moment and shear force at the mid point of a portal frame experimentally and compare them with theoretical values. b) To evaluate the lateral deflection of the portal frame under given loading conditions experimentally and verify it with theoretical predictions.
9.	Buckling of beam	Determine critical buckling load for fixed-fixed and pinned-pinned boundary conditions.
10.	Evaluation of forces and displacements in a redundant truss	a) To determine the forces in the members of a three-bar suspension system for vertical loads experimentally and verify the same with theoretical values. b) To determine the horizontal

		and vertical displacement of the loaded joint of a three-bar suspension system for vertical loads experimentally and verify the same with theoretical values.
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